AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Computer Science and Engineering

Course Title: Algorithm Lab Course Code: CSE 2208 Marks: 6

Assignment 2

Question: Suppose there is a large array of text sorted in ascending alphabetical order. Your task is to search for a list of words within this array using both **Linear Search** and **Binary Search**. Finally, you must compare the performance and efficiency of these two algorithms.

Input: There are two input files:

1. `searchspace.txt`: Contains approximately 100,000 words, representing the search space.
2. `wordspace.txt`: Contains approximately 100 words to be searched for within the search space.

The first line of each file specifies the total number of elements contained in that file. You must read the input from these two files.

Output: Run the Linear Search and Binary Search algorithms separately. Compare their performance based on the time and efficiency taken to locate the words within the search space.

Hint: Please use a compare or suitable way to compare string/words.